Michael Schlichtig

PhD Candidate (Computer Science)
Secure Software Engineering · Paderborn University · Heinz Nixdorf Institute

michael.schlichtig@uni-paderborn.de | +49 5251 60-6580 | Fürstenallee 11, 33102 Paderborn

schlichtig.org | github.com/schlichtig | linkedin.com/in/schlichtig

orcid.org/0000-0001-6600-6171 | scholar.google.com

Skills: Java, Python, C++ · Static Analysis (Soot, SootUp, CogniCrypt) · Android

Research Interests

Usability of static code analysis tools • Software API security and cryptographic library misuse • AI-assisted repair of security defects (LLMs + static analysis) • Software engineering with and for AI-based systems • Android privacy and GDPR compliance automation • Developer tooling (IDE plugins, static analysis frameworks)

Professional Experience

Research Associate, Heinz Nixdorf Institute **09/2019 – 09/2026**

Paderborn University, "Secure Software Engineering" group
(until 08/2019: "Didactics of Computer Science") – combined with doctoral studies

- CRC 1119 CROSSING, subproject E1 "Secure Integration of Cryptographic Software" (DFG) – Maintainer of CogniCrypt; SAST with Soot, SootUp, CI/CD integration
- Research on improving the usability of SAST tools
- AI-assisted repair of cryptographic API misuses (LLMs + SAST); SecAI tool
- Research on Android privacy and GDPR-compliant software development

Project Lead (Principal Investigator) **02/2023 – 08/2024**

Software Campus project "Specifiable automated detection of API misuse in CI pipelines" (API_ASSIST), BMBF – 100,000 EUR

Research Associate **08/2018 – 08/2019**

Paderborn University
"Didactics of Computer Science" group, project "Data Science and Big Data in Schools" (ProDaBi)

- Development and testing of a Data Science and AI curriculum for secondary school (age 16–18)

Selected Publications (peer-reviewed)

[1] M. Khedkar, M. Schlichtig, N. Atakishiyev, and E. Bodden. Between law and code: Challenges and opportunities for automating privacy assessments. *Automated Software Engineering*, 33(2):56, 2026.

[2] M. Khedkar, M. Schlichtig, M. Soliman, and E. Bodden. Challenges in Android data disclosure: An empirical study. In *Proceedings of the IEEE/ACM 13th International Conference on Mobile Software Engineering and Systems*. ACM, 2026.

[3] M. Nachtigall, M. Schlichtig, and E. Bodden. A large-scale study of usability criteria addressed by static analysis tools. In *Proceedings of the 31st ACM SIGSOFT International Symposium on Software Testing and Analysis*, pages 532–543, 2022.

[4] A. K. Wickert, M. Schlichtig, M. Vogel, L. Winter, M. Mezini, and E. Bodden. Supporting error chains in static analysis for precise evaluation results and enhanced usability. In *2024 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*, pages 693–704. IEEE, 2024.

[5] M. Schlichtig, S. Sassalla, K. Narasimhan, and E. Bodden. FUM – a framework for API usage constraint and misuse classification. In *Proceedings of the 2022 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*, pages 673–684, 2022.

Full list: schlichtig.org/publications | Google Scholar

Education

Doctorate in Computer Science **08/2018 – 08/2026**

Paderborn University, "Secure Software Engineering" group (until 08/2019 "Didactics of Computer Science")

Dissertation submitted; defense 31 August 2026

M.Sc. Computer Science (English-language program) **04/2017 – 08/2018**

Paderborn University

Overall grade 1.2 (with distinction); incl. semester abroad at University of Oklahoma (08–12/2015, GPA 3.75)

B.Sc. Computer Science **10/2011 – 03/2017**

Paderborn University – minor in Economics

Thesis: "Evaluation von Gestaltungsempfehlungen im Rahmen einer hypothesengeleiteten Technikgestaltung" (grade 1.0) – overall grade 2.2

Awards

CROSSING Collaboration Award **2021**

Collaborative Research Center (CRC) 1119 CROSSING, TU Darmstadt

Grants and Funding

API_ASSIST – "Specifiable automated detection of API misuse in CI pipelines" **02/2023 – 08/2024**

*100k EUR, BMBF Software Campus – **Principal Investigator***

Teaching and Supervision

Teaching Assistant (Course Support and Supervision) **Since 09/2019**

Paderborn University

- "Designing Code Analyses for Large-Scale Software Systems" (DECA) – master level
  Winter 2019, Winter 2020, Winter 2021, Winter 2023, Winter 2024
- "Secure Software Engineering" (SSE) – bachelor level
  Summer 2020
- "Software Engineering Internship" (SWTPra) – bachelor level
  Winter 2021, Winter 2022
- Project Group "AI-supported Security Testing" (SecAI) – one-year master's project, Winter 2024 – Summer 2025, independently conceived and led; SonarQube plugin for visualizing CogniCrypt analysis results with a focus on usability and AI assistance

Supervision of Theses, Student Assistants (SHK) and Research Assistants (WHB) **Since 08/2018**

Professional Service

Reviewer **2025 – 2026**

Journals

- *IEEE Transactions on Software Engineering*
- *ACM Computing Surveys*
- *The Journal of Systems & Software*

Program Committee Member **2025**

IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER '25) — Tool Demo Track

Research Visits

Research Visit **04/2025 – 05/2025**

Universidade de Brasília (UnB) and Universidade Federal de Pernambuco (UFPE), Brazil

- Host: Prof. Rodrigo Bonifácio, UnB, Brasília
- Invited talks: "Helping Developers Reduce Misuses of Java APIs" (UnB and UFPE)